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Experiment No 3

Title: Looping and strings

Aim: Write a program to demonstrate looping statements (while, for, nested loops, break, continue, else with loop), lists and string functions

Theory:

1) Differentiate between for loop and while loop with examples.

⇒ The for loop is used when the number of iterations is known in advance. It iterates over a sequence such as a list, string, or range object, and the loop variable takes each value from that sequence automatically. The while loop is used when the number of iterations depends on a condition. The loop body executes as long as the condition remains true, and the programmer must manually update the condition variable to avoid infinite execution.

Parameter	for loop	while loop
• Iteration control	over a sequence	Based on a condition
• Number of iterations	Known in advance	May be unknown
• counter update	Automatic	Manual
• Risk of infinite loop	Lower	Higher if condition not updated.

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Example of for loop:

code:
```
for i in range(1, 6):
    print("Iteration:", i)
```

output:
```
Iteration: 1
Iteration: 2
Iteration: 3
Iteration: 4
Iteration: 5
```

Example of while loop:

code:
```
i = 1
while i <= 5
    print("Iteration:", i)
    i += 1
```

output:
```
Iteration: 1
Iteration: 2
Iteration: 3
Iteration: 4
Iteration: 5
```

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2) Explain the role of the range() function in loops. Give two examples.

=> The range() function generates a sequence of integers and is commonly used with for loops to control the number of iterations. It accepts up to three arguments: start, stop, and step. The start value defaults to 0 and the step defaults to 1. The stop value is exclusive, meaning the sequence ends one integer before it.

Syntax:

```
range(stop)
range(start, stop)
range(start, stop, step)
```

Example 1:

~~for code:~~

```
for i in range(1,6)
    print("Iteration:", i)
```

output:

```
Iteration : 1
Iteration : 2
Iteration : 3
Iteration : 4
Iteration : 5
```

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Example 2: ~~Print~~

code:- for i in range(1,10,2)
print(i)

output:- 1 3 5 7 9

3) what is the purpose of the else block in a loop? Write one example using for loop with else.

=> The else block in a loop executes after the loop completes all its iterations normally. If the loop is terminated early by a break statement, the else block is skipped entirely. This makes it useful for detecting whether a loop ran to completion without interruption, such as in search operations where a break is used when the target is found.

code:- for i in range(1,6):
print(i)
else:
print("Loop completed without break")

output:- 1
2
3
4
5
Loop completed without break.

since no break was encountered, the else block is executed.

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4) Differentiate between break, continue, and pass with suitable examples.

=>

Keyword	Effect on loop	common use case
break	Exits the loop actually	stop execution when condition is met
continue	skips the current iteration	Bypass specific value in sequence
pass	no effect; loop proceeds normally	Placeholder where syntax requires a statement

Example of break:

```
for i in range(1, 10):
    if i == 5:
        break
    print(i, end=" ")
```

output: 1 2 3 4

The loop exits as soon as i reaches 5

Example of continue:

```
for i in range(1, 6):
    if i == 3:
        continue
    print(i, end=" ")
```

output: 1 2 4 5

The value 3 is skipped, all other iterations proceed normally.

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Example of pass:

```
for j in range(1,6):
    if j == 3:
        pass
    print(j, end=" ")
```

output: 1 2 3 4 5

5) Why are strings called immutable in Python? Explain with an example.

=> Strings in Python are called immutable because once a string object is created in memory, its individual characters cannot be modified in place. Any operation that appears to change a string does not alter the original object. Instead, Python creates a new string object with the modified content and assigns it to the variable. The original object remains unchanged.

This can be verified using the id() function, which returns the memory address of an object. After reassignment, the address changes, confirming that a new object was created.

code:

```
s = "hello"
print("Before:", id(s))

s = s + "world"
print("After:", id(s))
```

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output: Before: 140245678401234
After: 140245 67890 1235

The two different id values confirm that ~~assignment~~ reassignment created a new object rather than modifying the original string.

conclusion:-

This Experiment covered the core looping constructs in Python including for, while, nested loops, and loop control statement such as break, continue and pass. The behaviour of the else block with loops and the use of the range() function were also observed through code execution. String immutability was verified by attempting direct character modification and comparing object identifies using id().

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